THE NOVA RESEARCH CURRICULUM

Computer Science, Robotics, Finance, Research, Art, PE and Character

Grade 6 · Term 1

Student edition

Computer Science (3 lessons)

CS-6.3 (Weeks 1–8): input, processing, output and storage devices; system vs application software; files and folders; how the internet works (devices, routers, servers); strong passwords and two-step verification. CS-6.1 (Weeks 9–12): flowchart symbols (start/end, process, decision, input/output) and pseudocode for everyday algorithms.

#	Outcome	Question	Your answer
1	CS-6.3	A keyboard is an … device.	output / input / storage / processing
2	CS-6.3	An operating system is application software. True or false.	True / False
3	CS-6.3	Which is the strongest password?	123456 / ali2012 / Sky!River_74pine / password1
4	CS-6.1	In a flowchart, a decision is shown by a …	rectangle / oval / diamond / circle
5	CS-6.1	Write pseudocode to decide if a number is even.	_____

Robotics (1 lesson)

RO-6.1: build and explain a lever (ruler and eraser), gear trains (bottle caps), and a pulley (spool and string); record how each changes force or direction.

#	Question	Your answer
1	A pulley can change the direction of a force. True or false.	True / False
2	Two meshed gears turn in … directions.	the same / opposite

Financial Literacy (1 lesson)

FL-6.1: needs, wants and saving goals; a monthly budget for a family of four from a given income; review and adjust.

#	Question	Your answer
1	Income Rs 80,000; spending Rs 72,000. How much is saved, and what percentage?	_____

Research and projects (1 lesson)

RP-6.1: turn a topic into a question; judge sources (who wrote it, when, why, evidence); write a one-page summary with two references.

#	Question	Your answer
1	Which is a researchable question?	Pakistan / How does air pollution in Lahore change between summer and winter? / Pollution is bad / Air

Art, PE and Character

- **AR-6.1:** observational drawing, one- and two-point perspective, a portfolio piece.
- **PE-6.1:** football tactics (passing triangles, defending shape); fitness tracking; a team match.
- **CH-6.1:** a weekly planner, teamwork roles, and calm-down strategies for stress.

#	Outcome	Question	Your answer
1	AR-6.1	In one-point perspective, lines meet at a …	vanishing point / horizon only / corner / circle
2	CH-6.1	Breaking a big task into small steps helps manage time. True or false.	True / False